Bayesian Search — Streamlit App User Guide

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Live app: `appapppy-femambu7sdfwgwutujck7f.streamlit.app`

1 What the app does

Interactive companion to Deliverable 2. It lets you (i) inspect each of the twelve priors over the unknown cell Z, (ii) inspect the four detection models $\rho_j(d, r, e)$, (iii) preview what each of the four search strategies would propose given the current prior and detector, (iv) run a full interactive search campaign against a planted target, and (v) view the final operational pipeline applied end-to-end. All maps are computed live; no MCMC is required at runtime (Laplace / cell-level update only).

2 Sidebar — global configuration

Four controls live in the left sidebar and apply across all tabs:

Prior	one of the twelve registered priors P3–P14. Default is the operational pick `P13_impact_robust_uniform10`.
Detector	one of the four detector families D1–D4. Default is `D1_saturating_exponential`.
Update temperature	$T \geq 1$ in $\pi_{\text{post}}(j) \propto \pi_{\text{prior}}(j)\, L_j^T$. $T = 1$ is the textbook Bayes rule; $T = 1.5$ amplifies the partial-failure signal (operational default).
Budget cap	total cost allowed across the campaign (max 530).

3 The six tabs

1. Overview. Plots the accident point x_E, the velocity vectors $(v_{\text{plane}}, v_{\text{wind}}, v_{\text{drift}})$, the expected impact $\mu(\tau_{\text{fall}}, \tau_{\text{drift}})$ and the trajectory axes $(\hat{d}, \hat{n})$. Useful to build intuition before playing with priors.

2. Priors. Heatmap of the selected prior plus a 4×3 grid of all twelve priors at the same colour scale — the place to compare how much each prior concentrates around μ versus spreading laterally.

3. Detectors. Heatmap of ρ_j for the selected detector at the chosen effort $e \in \{1, 2, 3\}$, plus the four detector families side by side at the same effort.

4. Strategy preview. For the current (prior, detector) it shows what each of the four strategies (FIXED, ADAPTIVE, INFO_GAIN, PROGRESSIVE SCENARIO SEARCH) would propose as the next rectangle and effort. No mission is executed.

5. Simulator. The main interactive page. Plant a target (manual cell or random), then either fire a mission manually (rectangle + effort) or let the selected strategy propose one and execute. The outcome $s_t \in \{0, 1\}$ is drawn from the true detector, the cell-level posterior is updated with temperature T, the heatmap refreshes, and the mission is appended to the on-screen history table. Budget and entropy counters update live.

6. Operational model. The final delivered pipeline executed end-to-end with no user input: prior `P13`, detector `D1_saturating_exponential`, strategy *progressive scenario search* with the `mixture` hypothesis, budget $B = 530$. Shows the full mission sequence under repeated non-detection (the `missions.csv` export), the final posterior, and the per-mission posterior evolution.

4 Typical walkthroughs

To compare priors. Tab 2, eyeball the 4×3 grid, switch the sidebar prior to see the selected one larger.

To compare strategies on a single state. Tab 4 with a fixed sidebar (prior, detector).

To run a complete campaign by hand. Tab 5, plant a target, pick the strategy you want from the dropdown inside the tab, click *run mission* repeatedly until the budget is exhausted or the target is found.

To reproduce the deliverable. Tab 6, no clicks needed — it re-executes the same configuration that produced `missions.csv`.

5 Reset and caching

All priors and detector maps are cached (`st.cache_data`). If you change the sidebar values the affected tabs recompute; the others reuse the cache. To clear simulator state (planted target, mission history), press *Reset* at the top of tab 5.